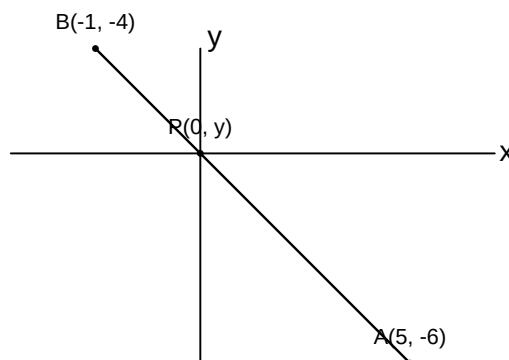


Coordinate Geometry Test — Solutions

1. Find the ratio in which the y -axis divides the line segment joining the points $A(5, -6)$ and $B(-1, -4)$. Also, find the point of intersection.



Answer:

Ratio is 5 : 1 and the point of intersection is $(0, -13/3)$.

Solution:

- Let the ratio in which the y -axis divides the segment AB be $k : 1$. Since the point lies on the y -axis, its x -coordinate must be 0.
- Apply the section formula for the x -coordinate: $x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}$. Substitute $x = 0$, $x_1 = 5$, $x_2 = -1$ to solve for k .
- Once k is found, use the section formula for the y -coordinate: $y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$ to find the intersection point.

Derivation:

$$0 = \frac{k(-1) + 1(5)}{k + 1}$$

$$\Rightarrow 0 = -k + 5$$

$$\Rightarrow k = 5$$

The ratio is 5 : 1

$$y = \frac{5(-4) + 1(-6)}{5 + 1} = \frac{-20 - 6}{6} = \frac{-26}{6} = -\frac{13}{3}$$

Point of intersection is $(0, -\frac{13}{3})$

Coordinate Geometry — Section Formula